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June 2004

INTERNATIONAL GCSE

MARK SCHEME

MAXIMUM MARK: 40

SYLLABUS/COMPONENT: 0625/06

PHYSICS
Alternative to Practical

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1	**(a)**	0.63 – 0.65 (A) (strictly)	
		1.64 – 1.66 (V) (strictly)	1
		3.32 (g)	1
		150 (cm^3)	1
		8 (mm) or 0.8 (cm)	1
		All units correct	1
	(b)	Remove electrodes from beaker	1
		A method to ensure gap remains the same (or other suitable suggestion e.g. measurement arrangement that the beaker sits on)	1
	(c)	New variable (e.g. temperature, surface area / vol / size of electrodes, power source setting, depth of immersion)	1
		TOTAL	**9**
2	**(a)**	All T values correct (0.34, 0.44, 0,49, 0.53, 0.60, 0.63)	1
		All T values to 2 sf OR all to 3sf	1
	(b)	Graph:	
		Scales suitable	1
		Scales labeled and with units	1
		Plots correct to ½ sq (-1 each error)	2
		Line judgement	1
		Line thickness (and small, neat plots)	1
	(c)	T = 0.51 (s) correct answer only; NO ecf	1
	(d)	Statement: NO	1
		Reason: line not through origin (or equivalent)	1
		(allow mark if candidate describes str. line or constant gradient)	
		TOTAL	**11**
3	**(a)**	Correct voltmeter	1
		Correct ammeter	1
	(b)	R = 3.3, 2/3 sf	1
		Unit Ω or ohm	1
	(c)	Circuit with correct parallel connections	1
		Ammeter and ONE voltmeter correct	1
		Variable resistor correct	1
		TOTAL	**7**

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4	**(a)**	**(i)**	x = 14 – 16mm	
		(ii)	y = 76.5 – 78.5 mm	1
		(iii)	u = 75mm (ecf) and v = 390mm (ecf)	1
			x,y,u and v all correct and with no unit	1
		(iv)	m = 5.2 (ecf) 2/3 sf and with no unit	1
	(b)		Upside down	1
			Precaution 1	1
			Precaution 2	1
			(e.g. repeats, use mark on block supporting lens to show centre of lens, place metre rule on bench to take readings or clamp rule in position, use a dark area, explanation of how to avoid parallax error, vertical screen/lens/both, centres of lens and object in line)	
			TOTAL	**8**
5	**(a)**		22	1
	(b)	**(i)**	14 (ecf)	1
		(ii)	64	1
			units all correct	1
	(c)		So that heat is not lost (wtte)	1
			TOTAL	**5**

PAPER TOTAL = [40]